

DR Tulu: Reinforcement Learning with Evolving Rubrics for Deep Research

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Outline

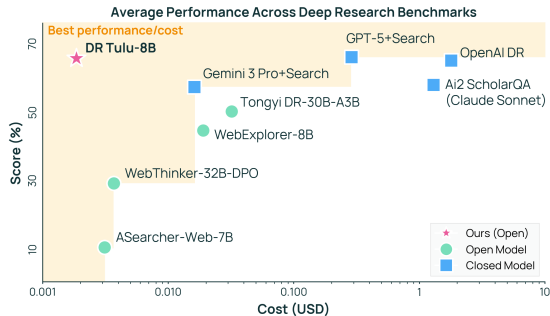
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1. Contribution: The Problem

- **Deep research (DR) agents** “produce in-depth, well-attributed answers to complex research tasks by planning, searching, and synthesizing information from diverse sources”
- Existing open DR agents are either:
 - **training-free**: manually designed prompts with off-the-shelf models, or
 - trained via **RLVR on short-form QA** – “does not extend to realistic long-form tasks”
- RL for open-ended DR critically depends on **reliable reward signals**, but:
 - desiderata for good responses are **under-specified** $\Rightarrow$ hard to capture with static, pre-defined evaluation criteria
 - accurate assessment requires **up-to-date external information** beyond a model’s parametric knowledge

1. Contribution

- **RLER**: rubrics constructed and maintained to *co-evolve* with the policy model during training
- **DR Tulu-8B**: first open model trained *end-to-end* for open-ended, long-form deep research
- Beats the strongest open 8–32B models (+**15.6** avg over Tongyi DR-30B); matches or exceeds OpenAI DR at $\sim 1000\times$ lower cost per query
- Full release: models, data, code, and `dr-agent-lib` (MCP-based multi-tool agent infrastructure)

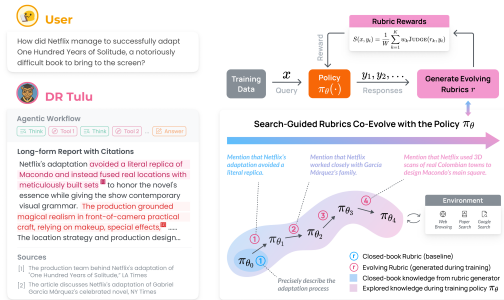


Avg. over 4 long-form benchmarks vs. cost:

DR Tulu-8B is on the Pareto frontier

2. Problem Formulation

- A **deep research model** = a language model equipped with search tools; each tool returns textual resources that can be **cited** in the answer
- Action space {think, call_tool, answer, cite}: sample actions autoregressively; tool outputs are appended to context; stop when the model chooses answer
- cite wraps claims and points to supporting snippets — “as localized as possible”



Left: example question and cited long-form response. Right: evolving rubrics vs. static closed-book rubrics.

2. Problem Formulation: Formal Definition

- **Tools** $\mathcal{T} = \{T_1, T_2, \dots\}$: each T_k takes a query q with optional arguments α , and returns a textual observation $o = T_k(q; \alpha)$
- **Policy** π_θ operates autoregressively over a text sequence s , initialized as $s_0 = x$ (the task and system instructions)
- At each step i , sample an action and its content/arguments: $(a_i, \zeta_i) \sim \pi_\theta(\cdot \mid s_i)$

$$s_{i+1} = \begin{cases} s_i \oplus \langle a_i, \zeta_i \rangle, & a_i \in \{\text{think}, \text{answer}, \text{cite}\} \\ s_i \oplus \langle a_i, \zeta_i, o_i \rangle, & o_i = T_k(q_i; \alpha_i), \quad a_i = \text{call_tool} \end{cases}$$

- Process continues until $a_\tau = \text{answer}$; ζ_τ contains the final answer
- Tool choice and arguments are set inside the protocol token, e.g.
`<call_tool name="google_search" k="10" lang="en">query</call_tool>`
- cite wraps claims in the answer with IDs of returned snippets — ideally “as localized as possible” (e.g., a snippet within a webpage vs. the entire webpage)

3. Rubrics as Rewards

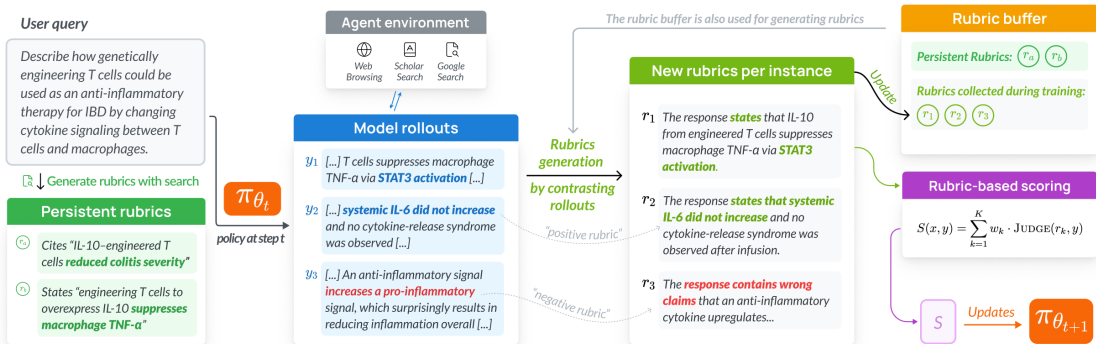
- **Rubrics** = explicit evaluation criteria for assessing (typically long-form) model responses
- **Sample-wise**: each question x has its own rubric set $\mathcal{R}_x = \{(r_{x,k}, w_{x,k})\}_{k=1}^K$ (criterion + weight)

$$S(x, y) = \frac{\sum_{k=1}^K w_{x,k} \text{JUDGE}(r_{x,k}, y)}{\sum_{k: w_{x,k} > 0} w_{x,k}}, \quad \text{JUDGE}(\cdot) \in \{0, 0.5, 1\}$$

- An LM judge scores the final response against each rubric item; RL maximizes the expected rubric score over training questions
- Why rubrics? “their concrete, well-defined items reduce susceptibility to judge model bias and promote objective evaluation”
- **Limitations of existing approaches:**
 - rely on human experts to write and refine rubrics, or assume reference answers
 - rubrics are **static**: “they do not adapt as the policy explores new evidence or behaviors”

4. Reinforcement Learning with Evolving Rubrics (RLER)

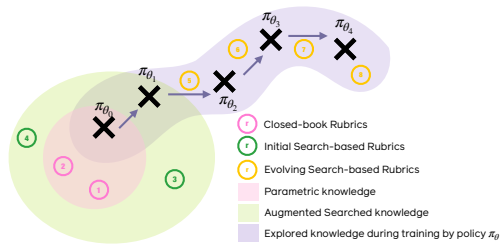
- Rubrics that are **instance-specific**, **grounded in external knowledge**, and **co-evolve with the policy model**



Sample rollouts $\rightarrow$ rubric LM proposes new rubrics from rollouts + current buffer $\rightarrow$ score rollouts and update policy $\rightarrow$ add and prune rubrics (keep highest rollout-score variance)

4.1 Key Intuition and Two Forms of Privileged Information

- Give the rubric generator **privileged information** “unavailable to the policy during generation, thereby creating a **generation–verification gap**”
- Two forms:
 - 1 **External knowledge** retrieved from multiple search rollouts $\rightarrow$ supports *fact verification*
 - 2 **Multiple independently sampled responses** $\rightarrow$ *contrastive signals* for assessing relative quality
- A new axis for scaling verifier compute: enrich the *information* available to the verifier, not just its reasoning tokens



Knowledge coverage: closed-book $\subset$ initial search-based
 $\subset$ evolving search-based rubrics

4.2 Initial Search-Based Rubrics

- Before training, for each prompt x :
 - run $\text{SEARCH}(x)$ via web search API
 - feed retrieved documents + question into rubric generator $\mathcal{G}_{\text{rubric}}$
- Produces **persistent rubrics** $\mathcal{R}_x^{\text{persist}}$, “persistently used throughout RL training”
- Search grounding makes rubrics **assertive** (“should mention benchmarks A and B”) rather than **descriptive** (“should discuss benchmarks”)
- Descriptive rubrics are gameable: score well “by superficially mentioning a point or even hallucinating facts”

Rubric type	Search	Assertive claims	
		Frac.	Factuality
General	×	0	–
Closed-book	×	0.22	0.94
Initial (search)	✓	0.56	0.97
Evolving	✓	0.52	1.00

Search-grounded rubrics are far more concrete, and their factual claims are almost always correct

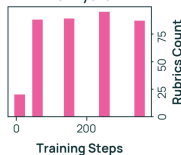
4.3 Evolving Rubrics During Training

- At each step, for prompt x and its G rollouts:
 - input = x + **all sampled responses** (“including the search context and final answers”) + existing rubric pool
 - $\mathcal{R}_x^{\text{new}} = \mathcal{G}_{\text{rubric}}(x, \{y_i\}_{i=1}^G, \mathcal{R}_x)$
- Two types of evolving rubrics:
 - Positive**: “strengths or new, relevant knowledge explored by the current policy” not yet in the pool
 - Negative**: “common undesirable behaviors, such as reward hacking” — e.g., verbatim copying of retrieved content to maximize citation precision

Proportion of Python Responses in Training



Negative Rubrics with Python



Negative evolving rubrics suppress an unexpected behavior (answering in Python code) that a static rubric could not anticipate

4.4 Rubric Buffer Management

- Without management, the number of rubrics “would grow linearly during training”
- After every GRPO rollout, score all G responses with the active rubrics:
 - **remove** rubrics with zero score variance — “they offer no discriminative value”
 - **rank** the rest by standard deviation; keep only the top- $K_{\max}$ ($K_{\max} = 5$)
- The buffer keeps whichever rubrics best *discriminate among current policy outputs*
- Plus small-weight auxiliary rewards: **format**, **search**, and **citation** (claim-level F1 from an LM judge)

4.4 Algorithm

Algorithm 1 Reinforcement Learning with Evolving Rubrics (RLER)

Require: Dataset $\mathcal{D}$, policy π_θ , rollout size G , max active rubrics $K_{\max}$, rubric generator $\mathcal{G}_{\text{rubric}}$

- 1: **for** each prompt $x \in \mathcal{D}$ **do**
 - 2: Generate $\mathcal{R}_x^{\text{persist}} \leftarrow \mathcal{G}_{\text{rubric}}(x, \text{SEARCH}(x))$ ▷ Generate initial search-based rubrics
 - 3: $\mathcal{R}_x^{\text{active}} \leftarrow \emptyset$
 - 4: **end for**
 - 5: **for** each training step $t = 1, \dots, T$ **do**
 - 6: $\mathcal{R}_x \leftarrow \mathcal{R}_x^{\text{persist}} \cup \mathcal{R}_x^{\text{active}}$
 - 7: Rollout with search $\{y_i\}_{i=1}^G \sim \pi_\theta(\cdot|x)$
 - 8: Generate $\mathcal{R}_x^{\text{new}} \leftarrow \mathcal{G}_{\text{rubric}}(x, \{y_i\}_{i=1}^G, \mathcal{R}_x)$; ▷ Generate evolving rubrics by contrasting rollouts
 - 9: $\mathcal{R}_x^{\text{active}} \leftarrow \mathcal{R}_x^{\text{new}} \cup \mathcal{R}_x^{\text{active}}$;
 - 10: Compute rewards with $\mathcal{R}_x^{\text{persist}} \cup \mathcal{R}_x^{\text{active}}$ and update π_θ (GRPO)
 - 11: Compute std of the rewards per rubric
 - 12: For $\mathcal{R}_x^{\text{active}}$, remove rubrics with 0 std; keep top- $K_{\max}$ with highest std ▷ Manage rubric buffer
 - 13: **end for**
-

5. DR Tulu with Open Infrastructure and Training Recipe

- Base model: **Qwen3-8B**
- Recipe:
 - ① `dr-agent-lib`: extensible multi-tool agent infrastructure (5.1)
 - ② **Supervised fine-tuning** as cold start on teacher trajectories (5.2)
 - ③ **Online RL** with RLER and asynchronous tool calls (5.3)
- Rubric generator: GPT-4.1 LM judge: GPT-4.1-mini

5.1 Agent Infrastructure: dr-agent-lib

- (i) Unified **MCP-based tool backend** integrating local and API-based search and browsing tools
- (ii) **High-concurrency backend**: global caching + asynchronous process locking for efficient, rate-limit-aware tool execution
- (iii) Lightweight, composable **prompt layer** for fine-grained control over search workflows

Tool	Backend	Returns
google_search	Serper API	top-10 web snippets
web_browse	Crawl4AI (train) / Jina (eval)	page text, summarized by Qwen3-8B
paper_search	Semantic Scholar full-text API	top-10 paragraphs from papers

Auto-search workflow; up to 10 tool calls per rollout; one unified pipeline for all tasks

5.2 Supervised Fine-Tuning for Cold Start

- Goal: “distill common search patterns from a teacher model” → better early rollouts, faster RL
- **Prompts:**
 - long-form: real user queries from SearchArena (general) + OpenScholar (scientific); LM prompt-filtering (1–5 quality score)
 - short-form: HotpotQA, TaskCraft, WebWalker-Silver, MegaScience + challenging synthetic prompts
- **Teacher trajectories:** GPT-5 generates reasoning + tool use + final answer under the auto-search workflow
- **Rejection sampling:** (i) keep only trajectories with correct tool-calling/answer format; (ii) for short-form, discard answers that do not match gold
- Yield: **16K SFT trajectories**; 5 epochs on one 8×H100 node (136 GPU-hours)

5.3 Online RL with Asynchronous Tool Calls

- Customized **GRPO** (Open-Instruct) with:
 - token-level loss (DAPO-style)
 - 1-step **asynchronous training**
 - **tool output masking**; sample packing
 - small KL penalty (0.001)
- **Asynchronous tool calls**: dispatched immediately during rollout; the request sleeps while the engine generates other responses
- RL prompts (long-form only): $\sim 5\text{K}$ SearchArena + OpenScholar, 4K RaR
- Scale: $16 \times \text{H100}$, **4,000 steps**, 27K GPU-hours, 70 days



Reward, output length, and tool calls during RL; training continued to improve even through periods of tool-API failures

6. Experimental Setup

- **Four long-form DR benchmarks** — all “require long-form responses and are evaluated using human-written or human-verified rubrics following official protocols”:
 - **SQAv2** (AstaBench-ScholarQA-CS2): scientific literature synthesis; rubric, answer, and citation precision/recall
 - **HealthBench**: healthcare; physician-written weighted rubrics (scores can be negative)
 - **ResearchQA**: research questions across fields
 - **DeepResearchBench (DRB)**: general-domain; comprehensiveness, depth, instruction-following, readability
- Baselines: closed DR systems, naive RAG, open DR models, fixed-pipeline DR
- DR Tulu: **one unified inference pipeline**, no task-specific customization; **snippet-level citations**

6. Main Results

	SQAv2	HealthBench	ResearchQA	DRB	Avg.
<i>Closed deep research</i>					
Perplexity Deep Research	67.3	—	75.3*	42.3*	—
Gemini 3 Pro + Search	69.8	38.0	74.3	46.3	57.0
GPT-5 + Our Search	61.1	31.1	62.8	50.3	51.3
GPT-5 + Search	74.8	59.5 [†]	78.2 [†]	50.7	65.8
OpenAI Deep Research	79.6	53.8 [†]	79.2 [†]	46.9*	64.9
<i>Naive RAG</i>					
Qwen3-8B	40.4	16.5	56.1	33.3	36.5
QwQ-32B	41.9	24.5	60.9	40.3	41.9
<i>Open deep research models</i>					
Search-R1-7B	22.2	-0.1	27.9	9.5	14.9
ASearcher-Web-7B	26.9	-13.0	19.4	7.8	10.3
WebExplorer-8B	42.5	33.7	64.8	36.7	44.4
WebThinker-32B-DPO	32.9	11.1	48.6	23.3	28.9
Tongyi DeepResearch-30B-A3B	46.5	46.2	66.7	40.6	50.0
<i>Fixed-pipeline deep research</i>					
WebThinker-32B-DPO (report)	46.7	39.4	74.2	40.6	50.2
Ai2 ScholarQA (Claude Sonnet)	87.7	32.0 [†]	75.0 [†]	36.1	57.7
<i>Open deep research (ours)</i>					
Qwen3-8B + Our Search	57.2	5.9	46.3	18.2	31.9
DR Tulu-8B (SFT)	72.3	38.1	68.5	39.0	53.9
DR Tulu-8B (RL)	88.3	52.8	75.7	45.4	65.6

*: reported by benchmark authors. †: 100-sample subset. Gray rows: closed backbones. Open baselines emit no citations ⇒ SQAv2 citation components are 0.

Three leaderboard-only rows omitted.

6. Performance Breakdown

	AstaBench-ScholarQA-CS2 (SQA v2)				DeepResearchBench (DRB)			
	Rubric	Answer	Cite-P	Cite-R	Comp	Depth	Instruction	Readability
<i>Closed Deep Research</i>								
🌟 Claude-Sonnet Search	-	-	-	-	39.0	37.7	45.8	41.5
🔍 Perplexity Sonar	-	-	-	-	37.4	36.1	45.7	44.7
🔍 Perplexity DR	91.6	92.7	47.3	37.6	40.7	39.3	46.4	44.3
🔍 Gemini Deep Research	-	-	-	-	48.5	48.5	49.2	49.4
🔍 Gemini3 Pro + Search	83.1	98.3	68.5	29.4	43.4	44.9	49.8	49.0
🔍 GPT-5 + Search	92.3	93.8	67.8	45.6	49.7	51.5	51.6	48.5
🔍 GPT-5 + Our Search	74.9	93.2	42.5	33.7	26.7	21.3	41.0	29.4
🔍 OpenAI DR	91.5	95.6	77.4	43.1	46.8	45.2	49.2	47.1
<i>Naive RAG</i>								
Qwen3-8B	69.2	92.3	-	-	29.4	27.0	40.2	41.1
QwQ-32B	77.5	90.3	-	-	38.1	34.8	47.0	44.6
<i>Open Deep Research</i>								
Search-R1-7B	9.7	79.0	-	-	5.2	2.1	18.6	16.8
ASearcher-7B	13.7	94.0	-	-	5.1	1.7	15.2	11.8
WebExplorer-8B	78.6	91.4	-	-	33.7	28.5	45.7	42.2
WebThinker-32B-DPO	36.7	94.9	-	-	19.7	12.3	36.8	26.3
Tongyi DeepResearch-30B-A3B	89.5	96.4	-	-	39.1	34.3	46.8	45.4
<i>Fixed Pipeline Deep Research</i>								
WebThinker QwQ-32B (report)	86.4	94.3	-	-	36.2	32.6	43.2	42.9
WebThinker-32B-DPO (report)	91.2	95.5	-	-	39.4	35.4	46.0	43.5
🔍 Ai2 ScholarQA - Claude Sonnet	88.1	89.1	92.4	81.2	35.1	32.0	40.5	38.9
<i>Open Deep Research (Ours)</i>								
Qwen3-8B + Our Search	42.8	92.1	53.7	40.3	14.3	8.7	29.5	24.4
🔍 DR Tulu-8B (SFT)	81.4	91.0	65.3	51.6	36.3	35.3	45.5	39.5
🔍 DR Tulu-8B (RL)	92.4	98.8	90.5	71.6	44.2	44.5	49.4	42.4

Table 11. Performance breakdown for Asta-ScholarQA-CS2 and DeepResearchBench. Open deep research models and naive RAG baselines do not provide citations, indicated as “-” in citation columns. Rows with a gray background indicate models that use closed models as backbone LMs. **Bold** indicates the best results among the baselines that do not use propriety models.

6. Main Results: Takeaways

- **Best open DR model:** +15.6 avg over the best prior open baseline (Tongyi DR-30B), using a 4× smaller dense model
- **Matches or exceeds proprietary systems:** strongest of all systems on SQA_{v2}; competitive with OpenAI DR on average (65.6 vs. 64.9)
- Models trained on short-form QA (Search-R1, ASearcher) “perform poorly on realistic report-length generation”
- Base model capability matters: GPT-5 + Search outperforms OpenAI DR on some datasets
- RLER adds **+6.4 to +16.0** points over SFT across the four benchmarks, including +25.2 citation precision and +20.0 citation recall on SQA_{v2}

6. Cost Efficiency

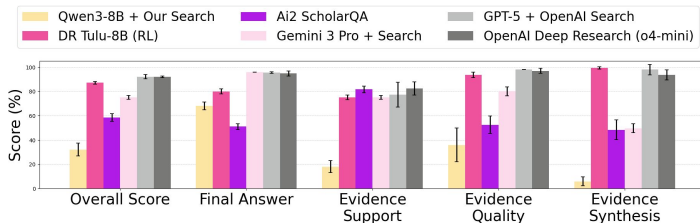
	Ans. len	Cites	Tool calls	\$/query
GPT-5 + Search	2358.7	28.1	–	0.29
OpenAI Deep Research	6445.1	79.6	–	1.80
Gemini 3 Pro + Search	1310.9	8.6	8.5	0.13
Ai2 ScholarQA	2090.5	61.2	1.0	1.30
Tongyi DR-30B-A3B	2138.9	–	23.0	0.032
WebThinker-32B (report)	4416.7	–	8.2	0.015
DR Tulu-8B (RL)	1889.2	35.8	4.3	0.0019

Usage statistics on SQA_{v2}

- **~1000× cheaper** than OpenAI DR per query (\$0.0019 vs. \$1.80)
- Efficiency comes from **adaptive tool usage**: free paper search dominates on SQA_{v2}
- Cheaper than all open DR baselines while scoring higher

6. Application: GeneticDiseasesQA

- 47 expert-curated questions covering 24 pathogenic gene variants (built with medical experts); models produce long-form, citation-backed reports
- Out-of-domain for DR Tulu — **no task-specific training**



DR Tulu-8B is competitive with proprietary systems on evidence support/quality/synthesis;

“no other open agents can tackle this task due to their inability to produce reliable, verifiable citations”

User

What is the pathomechanism of variant
NM_001127222.2(CACNA1A):c.4174G>A?

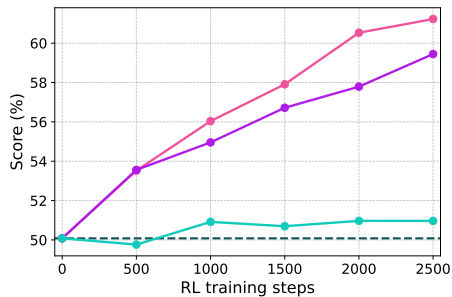
DR Tulu

CACNA1A variant c.4174G>A translates to the protein change p.Val1392Met [...] reported across EA2, FHM1, and SCA phenotypes ^[1] an LoF mechanism fits EA2 and some SCA forms, whereas a GoF or dominant-toxic mechanism fits migraine and certain neurodevelopmental/epileptic phenotypes ^[2]

Sources

^[1] Clinvar, RCV000415108 ^[2] PubMed Central, PMC11926052

6. Ablations: Evolving Rubrics and Rubric Types



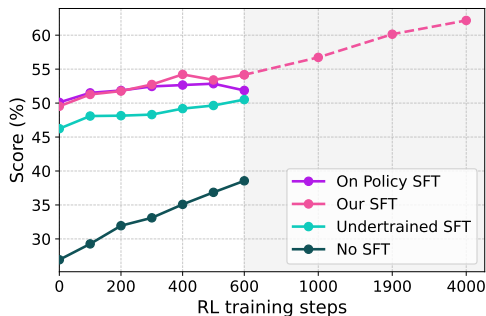
--- No RL ● w/ RLER ● w/ initial rubrics only ● w/ random

RLER > static search-based rubrics > random rewards / no RL;
the gap widens as evolving rubrics capture newly explored
knowledge

	SQAv2	Health	Res.QA	DRB	Avg.
SFT v0.1	73.4	37.5	68.6	39.4	54.7
General	80.6	36.0	65.0	37.5	54.8
Closed-book	83.2	34.8	66.6	37.6	55.6
Search-based	82.8	37.9	66.9	39.3	56.7

500 RL steps with *static* rubrics only: search grounding already
helps, even before adding rubric evolution

6. Ablations: SFT Cold Start and Judge Choice

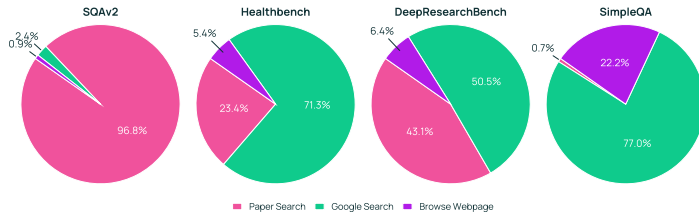


RL benefits from a stronger SFT start: even 5% of the SFT data
beats starting RL from the raw base model

	Avg. (4 benchmarks)
DR Tulu-8B (SFT)	53.9
+ RL 1k steps, GPT-4.1 judge	59.6
+ RL 1k steps, Qwen3-8B judge	58.3

RLER does not rely on a strong proprietary judge: using Qwen3-8B (the starting model itself) as judge and rubric generator still gains +4.4 over SFT, only 1.3 behind the GPT configuration

6. Analysis: Adaptive Tool Use and Short-Form Generalization



paper_search dominates on SQA v2 (~90%); web search/browse dominate on HealthBench, DRB, SimpleQA

	SimpleQA	2Wiki	WebWalker	Avg.
Qwen3-8B + Our Search	70.5	44.0	27.9	47.5
DR Tulu-8B (SFT)	75.5	66.5	31.9	58.0
DR Tulu-8B (RL)	75.9	68.9	39.0	61.3

RL uses *only long-form* prompts, yet short-form QA still improves (+3.3 avg) — strong cross-task generalization

- **Evolving rubrics adapt the verifier** to the policy's current behavior: on-policy, discriminative feedback for hard-to-verify long-form tasks
- **Privileged information** (retrieved knowledge + contrastive rollouts) is a new dimension for scaling verifier compute
- An **8B open model** can match proprietary deep research systems – at $\sim 1000\times$ lower cost
- Open challenge: **train–test mismatch** – the highest training reward does not always give the best downstream score
- Fully open release: models, data, training code, `dr-agent-lib`, evaluation suite